Ex. 32. ***A man spends $\frac{2}{5}$ of his salary on house rent, $\frac{3}{10}$ of his salary on food and $\frac{1}{8}$ of his salary on conveyance. If he has ₹ 1400 left with him, find his expenditure on food and conveyance.***

Sol. Part of the salary left $= 1-\left(\frac{2}{5}+\frac{3}{10}+\frac{1}{8}\right)=1-\frac{33}{40}=\frac{7}{40}$.

Let the monthly salary be ₹ x.

Then, $\frac{7}{40}$ of $x = 1400 \Leftrightarrow x=\left(\frac{1400\times 40}{7}\right)=8000.$

$\therefore$ Expenditure on food = ₹ $\left(\frac{3}{10}\times 8000\right)$ = ₹ 2400.

Expenditure on conveyance = ₹ $\left(\frac{1}{8}\times 8000\right)$ = ₹ 1000.

Ex. 33. ***A third of Arun's marks in Mathematics exceeds a half of his marks in English by 30. If he got 240 marks in the two subjects together, how many marks did he get in English?***

Sol. Let Arun's marks in Mathematics and English be x and y respectively.

Then, $\frac{1}{3}x-\frac{1}{2}y=30 \Leftrightarrow 2x-3y=180$ ...(*i*) and $x+y=240$ ...(*ii*)

Solving (*i*) and (*ii*), we get: $x = 180$ and $y = 60$.

$\therefore$ Arun's marks in English = 60.

Ex. 34. ***A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil were taken out and four bottles of oil were poured into it, it was $\frac{3}{4}$ full. How many bottles of oil can the tin contain?***

Sol. Suppose x bottles can fill the tin completely.

Then, $\frac{4}{5}x-\frac{3}{4}x=(6-4) \Leftrightarrow \frac{x}{20}=2 \Leftrightarrow x=40.$

$\therefore$ Required number of bottles = 40.

Ex. 35. ***If $\frac{1}{8}$ of a pencil is black, $\frac{1}{2}$ of the remaining is white and the remaining $3\frac{1}{2}$ cm is blue, find the total length of the pencil.***

Sol. Let the total length of the pencil be x cm. Then,

Black part $=\left(\frac{x}{8}\right)$cm. Remaining part $=\left(x-\frac{x}{8}\right)$cm $=\left(\frac{7x}{8}\right)$cm.

White part $=\left(\frac{1}{2}\times\frac{7x}{8}\right)$cm $=\left(\frac{7x}{16}\right)$cm. Remaining part $=\left(\frac{7x}{8}-\frac{7x}{16}\right)$cm $=\frac{7x}{16}$ cm.

$\therefore \quad \frac{7x}{16}=\frac{7}{2}$ or $x=\frac{16}{2}=8$ cm.

Hence, total length of the pencil = 8 cm.

Ex. 36. ***At a college football game, $\frac{4}{5}$ of the seats in the lower deck of the stadium were sold. If $\frac{1}{4}$ of all the seating in the stadium is located in the lower deck, and if $\frac{2}{3}$ of all the seats in the stadium were sold, what fraction of the unsold seats in the stadium was in the lower deck?***

Sol. Let the total number of seats in the stadium be x.

Then, number of seats in the lower deck $=\frac{x}{4}$.